

Max Hamilton

jmhamilton@umass.edu | Amherst, MA | johnmaxh.github.io

Research Profile

ML researcher developing statistically efficient model evaluation and human feedback methods, with a background in computer vision and generative models.

Research interests: Model evaluation; efficient human feedback; adaptive sampling and uncertainty quantification; generative models. Also interested in applying these methods to RLHF and post-training.

Education

University of Massachusetts, Amherst

Ph.D. in Computer Science

Advisor: Subhransu Maji

Sep 2022 - Dec 2026 (Expected)

University of Michigan, Ann Arbor

M.S.E. in Computer Science and Engineering

Cumulative GPA: 4.00/4.00

Sep 2020 - May 2022

University of Michigan, Ann Arbor

B.S.E. in Computer Science and Engineering

Minor in Mathematics

Cumulative GPA: 4.00/4.00

Sep 2016 - May 2020

Industry Research Experience

Adobe Inc., PhD Research Intern, San Jose, CA

Mentors: David Arbour, Branislav Kveton

June 2026 – Sep 2026

- Developed an efficient human evaluation framework for generative AI models, providing unbiased estimates of quality metrics with confidence intervals and statistical guarantees.
- Reduced human labeling effort by approximately 30% versus uniform sampling across four quality metrics on a generative video evaluation dataset.

Dolby Laboratories, PhD Research Intern, Sunnyvale, CA

Supervisor: Jaclyn Pytlarz

June 2024 – Sep 2024

- Cut diffusion-model inference cost 8x while matching state-of-the-art relative-depth accuracy, for monocular depth prediction.
- Designed efficient fine-tuning strategies for diffusion models on geometric vision tasks (monocular depth and surface normal estimation).

Skills

Programming Languages: Python, SQL, C++, MATLAB

Machine Learning: PyTorch, NumPy, Hugging Face (Transformers, Diffusers, Accelerate), TensorBoard, Optuna

Training & Scale: Mixed-precision training (AMP, fp16/bf16), AWS, GCP, Slurm

Academic Research Experience

Computer Vision Lab, Graduate Research Assistant, Amherst, MA

Sep 2022 - Present

Advisor: Prof. Subhransu Maji

- Developed *Active Measurement*, an adaptive sampling framework yielding statistically unbiased, high-precision aggregates from unlabeled data, enabling a 10x reduction in human labeling cost for quantifying malaria cells, satellite building damage, and bird populations. (NeurIPS 2025)
- Extended these unbiased estimators to compute Two-Point Correlation Functions (2PCF) for large-scale astronomy datasets. (AISTATS 2026)
- Developed a model-assisted multi-target estimation method that reduced average labeling effort 3x versus uniform-sampling baselines, validated across coral reef species, crop/land-use, and object counting datasets. (UAI 2026)
- Built a zero-shot species range estimation model. Engineered a scalable multimodal pipeline processing 37k+ Wikipedia articles into embeddings to train a novel text-to-range-map system using observational and textual data. (NeurIPS 2024)

Owens Research Group, Graduate Research Assistant, Ann Arbor, MI

Apr 2021 - Sep 2022

Advisor: Prof. Andrew Owens

- Achieved state-of-the-art video next frame prediction with correspondence-wise loss functions that match content rather than fixed pixel location. (CVPR 2022)
- Incorporated optical flow methods into the training objective to align predicted and ground-truth frames.

Fouhey AI Lab, Undergraduate Research Assistant, Ann Arbor, MI

May 2019 - Apr 2021

Advisor: Prof. David Fouhey

- Led a machine learning weather pattern localization project in PyTorch.
- Constructed a 10M+ geo-tagged sky image dataset to enable supervision at scale
- Explored self-supervised and implicit representations for image-based geo-localization

Vision and Learning Lab, Undergraduate Research Assistant, Ann Arbor, MI

May 2018 - Jan 2019

Advisor: Prof. Jia Deng

- Built a two-stage optimization pipeline in C++ to reconstruct 3D meshes from annotated 2D images, solving constrained curvature minimization for surface normal estimation
- Collected 140,000+ real-world object images via a custom web crawler, extracting camera pose from EXIF metadata. (contributed to OASIS, CVPR 2020)

Honors & Awards

Edward Riseman and Allen Hanson Scholarship, University of Massachusetts, Amherst

Apr 2023

Christopher P. Moeller Scholarship Award, University of Michigan

Sep 2021

Awarded to outstanding graduate students studying computer science or mathematics.

Publications

Scalable Model-Assisted Multi-Target Estimation in Large Image Collections

Max Hamilton, Subhransu Maji, Daniel Sheldon

Conference on Uncertainty in Artificial Intelligence (UAI), 2026.

Active Measurement of Two Point Correlations

Max Hamilton, Subhransu Maji, Daniel Sheldon

International Conference on Artificial Intelligence and Statistics (AISTATS), 2026.

Active Measurement: Efficient Estimation at Scale

Max Hamilton*, Jinlin Lai*, Wenlong Zhao, Subhransu Maji, Daniel Sheldon

Advances in Neural Information Processing Systems (NeurIPS), 2025.

Feedforward Few-shot Species Range Estimation

Christian Lange, **Max Hamilton**, Elijah Cole, Alexander Shepard, Samuel Heinrich, Angela Zhu, Subhransu Maji, Grant Van Horn, and Oisin Mac Aodha

Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.

Combining Observational Data and Language for Species Range Estimation

Max Hamilton, Christian Lange, Elijah Cole, Alexander Shepard, Samuel Heinrich, Oisin Mac Aodha, Grant Van Horn, and Subhransu Maji

Advances in Neural Information Processing Systems (NeurIPS), 2024.

Comparing Correspondences: Video Prediction with Correspondence-wise Losses

Daniel Geng, **Max Hamilton**, and Andrew Owens

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

OASIS: A Large-Scale Dataset for Single Image 3D in the Wild

Weifeng Chen, Shengyi Qian, David Fan, Noriyuki Kojima, **Max Hamilton**, Jia Deng

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

* Equal contribution.

Teaching Experience

University of Massachusetts Amherst, Teaching Assistant

CompSci 670: Computer Vision

Spring 2024

CompSci 682: Neural Networks

Fall 2023, Spring 2026

University of Michigan, Graduate Student Instructor

EECS 542: Advanced Topics in Computer Vision

Winter 2022

Professional Activities

Mentor, Early Research Scholars Program, University of Massachusetts, Amherst

2024 – 2026

Program committee, New England Computer Vision (NECV) Workshop

2025

Reviewer: ICCV 2023, WACV 2024/2025, ECCV 2024/2026, NeurIPS 2025, CVPR 2026